Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Python · Cloud-Native Systems · GenAI · AWS · Large-Scale Product Delivery

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Arlington area, Virginia

PROFESSIONAL SUMMARY

Senior software engineer with 8+ years progressing from Associate to Senior at Entrata — designing and delivering large-scale, cloud-native production systems serving **100K+ MAU across 1M+ records at 99.5% uptime**. Shipped **20+ production modules** (Python, JavaScript, TypeScript, SQL, REST APIs) across **120+ Agile sprints**; led Docker microservices containerization cutting execution time **50%**; migrated all client data to **AWS S3** with 100% PHI compliance, zero incidents. Extended into GenAI: built **AgentProbe** (3-agent LangGraph runtime, 15-endpoint FastAPI, LLM-as-judge eval engine, multi-provider routing) and **Leasel** (agentic RAG compliance pipeline, **93.6% precision**). Conducted 300+ peer code reviews, mentored 5+ engineers. M.S. Data Analytics Engineering, GMU (GPA 3.87).

WORK EXPERIENCE

Entrata

Jan 2021 – Aug 2024

Senior Software Engineer — Cloud-Native Architecture · GenAI Pipelines · Large-Scale Systems

- Architected Docker-based microservices containerization of the production platform — **cut execution time 50%**; migrated all client data to **AWS S3** with Libsodium-encrypted pipelines achieving **100% PHI/HIPAA compliance**, zero security incidents.
- Designed and shipped Entrata's first production AI pipeline across **1.6M real financial records** (Python, PyTorch, OpenAI API) with output validation, quality gates, and compliance controls at launch; later productized into a customer-facing feature.
- Delivered **20+ full-stack modules** (Python, JavaScript, TypeScript, REST APIs, CI/CD) across **120+ Agile sprints** — **100K+ MAU at 99.5% uptime**; conducted 300+ peer code reviews and mentored 5+ junior engineers on system design and production best practices.

Entrata

Jan 2019 – Dec 2020

Software Engineer — Automation Pipelines · Compliance Systems · Data Engineering

- Encoded manual income eligibility review logic into automated compliance pipelines (Python, PL/SQL) — AMI threshold calculations, recertification workflows, escalation routing across **millions of tenant certifications**; eliminated **80% of manual workload** with zero violations.
- Optimized **200+ SQL queries for 30% performance improvement** across 1M+ property records; built property performance dashboards tracking eligibility rates, occupancy, delinquency, and rent-vs-AMI trends.

Entrata

Jan 2017 – Dec 2018

Associate Engineer — Full-Stack Development · Property Management Systems

- Built and maintained full-stack property management features (PHP, JavaScript, SQL, REST APIs) across 100K+ unit portfolios — delivered across Agile sprints maintaining **99.5% production uptime** from day one.
- Developed compliance features for Section 8/HCV and LIHTC eligibility workflows; wrote and tuned SQL queries across all client database environments.

George Mason University

Aug 2024 – Aug 2025

Research Associate — ML Systems · Production Pipelines · Applied AI

- Built end-to-end ML pipelines (Python, R) on 190+ countries of WHO + UK Road Safety data — regression models achieving **$R^2 = 0.977$**; deployed Isolation Forest anomaly detection across 3 pipelines surfacing outliers manual review missed; shipped self-service dashboards (Tableau, ggplot2, Quarto) improving research comprehension **40%**.

PROJECTS

AgentProbe: Multi-Agent AI Testing Toolkit

Python · LangGraph · FastAPI · ChromaDB · Anthropic · OpenAI · Groq · Ollama · React · Docker · GitHub Actions — github.com/venkatalakshmiparimala/AgentProbe

- Built a **3-agent LangGraph runtime** (Orchestrator, RAG Tracer, Auto Fix) with a **15-endpoint FastAPI backend**, multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama), and **5-metric LLM-as-judge eval engine**; integrated CI/CD via GitHub Actions so every commit is automatically evaluated — runs 100% locally at zero cost.

Leasel: Agentic RAG Compliance Pipeline

Python · LangGraph · RAG · Scikit-learn · SQLite · reportlab — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Designed and shipped an end-to-end **LangGraph agentic workflow** processing 1,240 compliance documents — document extraction, NLP classification, and deterministic audit engine; **93.6% precision, 87.9% F1**; pass@1 81.7%, pass@3 88.3%; every determination fully auditable with metadata lineage.

CrashClarity: Large-Scale ML Classification (104K Records)

Python · Scikit-learn · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages — Live Dashboard

- Built Random Forest + Logistic Regression ensemble with end-to-end feature engineering across 104K+ UK accident records — **85%+ classification accuracy**; deployed interactive self-service dashboard on GitHub Pages.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia · ML systems, cloud-native pipelines, statistical modeling, AI systems, data engineering

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur — 2013 – 2017

TECHNICAL SKILLS

Languages & Frameworks: Python, Java, JavaScript, TypeScript, SQL, PL/SQL, R, PHP, Bash, HTML/CSS, FastAPI, REST APIs, NodeJS

GenAI & AI Systems: LangChain, LangGraph, Multi-Agent Orchestration, RAG Pipelines, LLM-as-Judge Eval, Prompt Engineering, OpenAI API, Anthropic API, Groq, Ollama, ChromaDB, BERT, spaCy, Scikit-learn, PyTorch, TensorFlow

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Docker, Microservices, CI/CD (GitHub Actions), DevSecOps, PostgreSQL, MySQL, Oracle, SQLite, ETL Pipelines, Linux, Git

Engineering Practices: Object-Oriented Programming (OOP), Design Patterns, System Design & Architecture, Agile/Scrum (120+ sprints), Peer Code Review (300+), DevSecOps, XP, Lean, PHI/HIPAA Compliance